BASIC FILE INPUT/OUTPUT

Basic file IO is quite easy to do.

Place a filename on top of the RTS. Then call the 'read-file' operator. The file will now be read and stored in a block of chars with '\0' to end it. The file is closed and so is the FILE descriptor.

We can grab this heap allocation into a pointer, like this:

```
/tmp/input-file.dat read-file data-ptr def-i
```

To open a file to write to, we use the open-file:

```
/tmp/file-output.dat open-file
"hello how are you?"
write-file
close-file
```

Open-file operator opens the file and gets it ready for the user to write to it.

write-file operator writes the value at the top of the RTS to the file placed below it.

There is only 1 file descriptor variable declared in .bss, write-fd. That means we can only ever open/read/write a single file at one time. write-fd is 8 bytes.

```
.bss write-fd resq 0x01
```

Correction to the previous page - Dec 2025

After I had written this page, I later added system-calls to this language. That means that we are also able to use the Sys-open system call #2 and grab the return value as an integer / pointer!

Now we're not restricted by a single file descriptor!